

Foundation (Jalapeno)

- Q1.** A Martian rover is located at coordinate -14 on a linear survey track. What is its distance from the landing base at coordinate 0 ?
- (A) -14
(B) 0
(C) 14
(D) 28
(E) -1
- Q2.** The temperature of a sample in a vacuum chamber drops to -78°C . The magnitude of this temperature is expressed as $|-78|$. Find the value.
- (A) -78
(B) 78
(C) 0
(D) 156
(E) -156
- Q3.** In a particle accelerator, a proton is at position $x = -5$ and an electron is at $y = 5$. Which statement is true regarding their absolute values?
- (A) $|x| < |y|$
(B) $|x| > |y|$
(C) $|x| = |y|$
(D) $|x| + |y| = 0$
(E) $|x| = -|y|$
- Q4.** Evaluate the expression for a satellite's orbital correction:
 $|-25| + |12| - |-3|$.
- (A) 34
(B) 40
(C) 10
(D) 37
- (E) 31
- Q5.** A lens in a telescope has a focus error of d . If $|d| = 0.005$ mm, what are the possible values for d ?
- (A) Only 0.005
(B) Only -0.005
(C) 0.005 and -0.005
(D) 0 and 0.005
(E) 0.010 and -0.010
- Q6.** Which of the following measurements has the smallest absolute value?
- (A) -12.5
(B) 10.2
(C) $|-9.8|$
(D) -5.1
(E) $|4.9|$
- Q7.** Calculate the total energy deviation in a lab experiment:
 $2 \cdot |-15| + |-4|$.
- (A) 26
(B) 34
(C) -26
(D) -34
(E) 30
- Q8.** A probe moves from -120 meters to $+30$ meters relative to a moon's surface. What is the absolute distance traveled?
- (A) 90 m
(B) -90 m
(C) 150 m
(D) -150 m
(E) 0 m

Open Ended Questions (FRQ)

- Q9.** A deep-space sensor records a primary reading of -42 units and a secondary reading of $+18$ units. Calculate the value of $|-42| - |18|$ and explain what the result represents in terms of distance from zero.
- Q10.** Evaluate the complex absolute value expression found in a rocket trajectory formula: $|-10 + 4| \cdot |2 - 8|$. Show each step of your calculation.

Chef's Tips

- Remind students that absolute value always results in a non-negative number.
- Use the number line to visualize distance from the origin (zero).
- Ensure students simplify the inside of absolute value bars before removing them.